

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics

PRACTICAL II

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Class: TYIT	Batch: 1
Date of Assignment: 31-07-2026	Date/Time of Submission: 31-07-2026

A) Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.

Code:-

```
import pandas as pd
df = pd.read_csv("1000_Companies.csv")
print(df.head())
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

```
sales = df["Profit"]
print("Mean :", sales.mean())
```

```
Mean : 119546.16465561
```

```
print("Mode :")
print(sales.mode())
```

```
Mode :
0    121248.7116
1    180257.1795
Name: Profit, dtype: float64
```

```
print("Minimum :", sales.min())
print("Maximum :", sales.max())
print("Range :", sales.max() - sales.min())
```

```
Minimum : 14681.4
Maximum : 476485.43
Range : 461804.02999999997
```

```
print("Variance :", sales.var())
print("Standard Deviation :", sales.std())
```

```
Variance : 1839434913.3887165
Standard Deviation : 42888.63384847688
```

```
print("Count :", sales.count())
print("Sum :", sales.sum())
```

```
Count : 1000
Sum : 119546164.65561
```

```
print("\nDescriptive Statistics")
print(sales.describe())
```

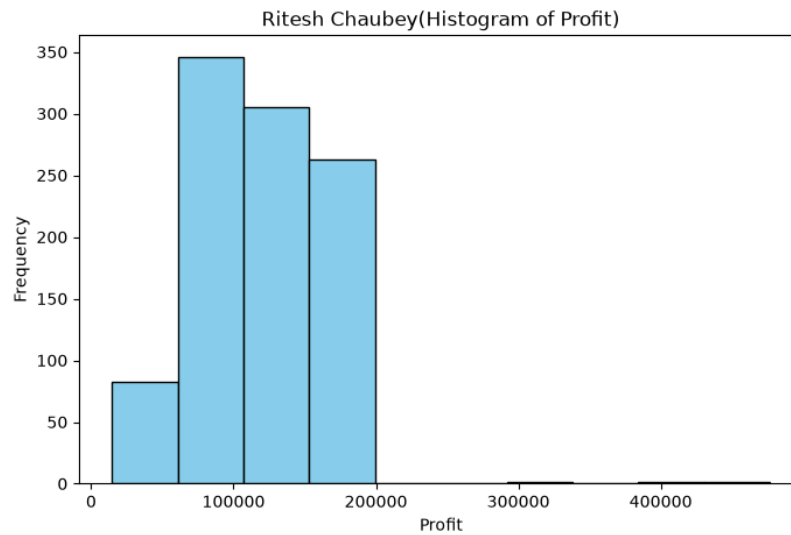
```
Descriptive Statistics
count      1000.000000
mean       119546.164656
std         42888.633848
min         14681.400000
25%         85943.198543
50%        117641.466300
75%        155577.107425
max         476485.430000
Name: Profit, dtype: float64
```

B) Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

Code & Output:-

Histogram of Profit

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8,5))
plt.hist(sales, bins=10, color="skyblue", edgecolor="black")
plt.title("Ritesh Chaubey(Histogram of Profit)")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
```



```
import seaborn as sns
plt.figure(figsize=(8,3))
sns.boxplot(x=sales, color="orange")
plt.title("Ritesh Chaubey(Box Plot of Profit)")
plt.xlabel("Profit")
plt.show()
```

